

- explanation of the **joint weighted fit**
- why the **R@highest O₂ reference** is used
- description of outputs and files

If you want, I can also expand it with:

- a **flow diagram of the calibration workflow**
- the **exact equations used in the MATLAB code**
- a **section explaining how to convert new optode images to O₂ maps** using the fitted parameters. `</>`



↳ Joint model used ----- The script uses: $R(C) = R_{high} \alpha + (1 - \alpha) / (1 + K_{SV} C) + (1 - \alpha) / (1 + K_{SV} C_{high})$
 $R(C) = R_{high} \frac{\alpha + (1 - \alpha) / (1 + K_{SV} C)}{\alpha + (1 - \alpha) / (1 + K_{SV} C_{high})}$
 $R(C) = R_{high} \alpha + (1 - \alpha) / (1 + K_{SV} C_{high}) \alpha + \dots$

prove how you get this equation?

Yes. It comes directly from the usual modified Stern–Volmer equation, just rewritten so the reference point is the **highest O₂** ratio instead of the **zero O₂** ratio.

I'll derive it step by step.

1. Start from the usual modified Stern–Volmer form

For one optode, the model is

$$R(C) = R_0 \left[\alpha + (1 - \alpha) \frac{1}{1 + K_{SV} C} \right]$$

where:

- $R(C)$ = ratio at oxygen concentration C
- R_0 = ratio at $C = 0$
- α = non-quenchable fraction
- K_{SV} = Stern–Volmer constant

Let me define

$$F(C) = \alpha + (1 - \alpha) \frac{1}{1 + K_{SV} C}$$

so the model becomes

$$R(C) = R_0 F(C)$$

That is the key simplification.

2. Write the same model at the highest oxygen calibration point

Let the highest oxygen concentration for optode f be

$$C_{\text{high},f}$$

and the corresponding measured ratio be

$$R_{\text{high},f}$$

Then, using the same model,

$$R_{\text{high},f} = R_0 F(C_{\text{high},f})$$

In expanded form:

$$R_{\text{high},f} = R_0 \left[\alpha + (1 - \alpha) \frac{1}{1 + K_{SV} C_{\text{high},f}} \right]$$

3. Solve this for R_0

From

$$R_{\text{high},f} = R_0 F(C_{\text{high},f})$$

we get

$$R_0 = \frac{R_{\text{high},f}}{F(C_{\text{high},f})}$$

or explicitly,

$$R_0 = \frac{R_{\text{high},f}}{\alpha + (1 - \alpha) \frac{1}{1 + K_{SV} C_{\text{high},f}}}$$

4. Substitute that back into the original equation

Original equation:

$$R(C) = R_0 F(C)$$

Substitute the expression for R_0 :

$$R(C) = \frac{R_{\text{high},f}}{F(C_{\text{high},f})} F(C)$$

So:

$$R(C) = R_{\text{high},f} \frac{F(C)}{F(C_{\text{high},f})}$$

Now replace $F(C)$ with its full definition:

$$R(C) = R_{\text{high},f} \frac{\alpha + (1 - \alpha) \frac{1}{1 + K_{SV} C}}{\alpha + (1 - \alpha) \frac{1}{1 + K_{SV} C_{\text{high},f}}}$$

That is exactly the equation used in the joint model.

5. Why this works

It is not a new physical model.

It is just the **same modified Stern–Volmer model**, but expressed using a different reference quantity:

- usual version uses R_0 , the ratio at zero oxygen
- rewritten version uses R_{high} , the ratio at the highest oxygen calibration point

So nothing fundamental changed. We only eliminated R_0 algebraically.

6. Quick check that the equation is correct

If you plug in $C = C_{\text{high},f}$, then numerator = denominator, so

$$R(C_{\text{high},f}) = R_{\text{high},f}$$

which is exactly what we want.

So the curve is anchored at the highest O₂ point.

7. Why this is useful for your microfluidic case

Your third optode starts from a **fully oxidic image**.

So it is much easier to know or estimate:

$$R_{\text{high,new}}$$

than to know:

$$R_0$$

for that specific device.

That is why the reparameterized form is practical.

8. One-line summary

The equation

$$R(C) = R_{\text{high},f} \frac{\alpha + (1 - \alpha)/(1 + K_{SV}C)}{\alpha + (1 - \alpha)/(1 + K_{SV}C_{\text{high},f})}$$

is obtained by:

1. starting from the standard modified Stern–Volmer equation,
2. evaluating it at the highest oxygen point,
3. solving for R_0 ,
4. substituting back.

If you want, I can also derive the **inverse equation**, meaning how to solve this expression for C so you can convert ratio images into oxygen concentration.

